PROJECT REPORT

"3D AR-Navigation"

A Project Synopsis submitted in partial fulfilment of the requirements for the award of the degree of

Bachelor of Technology (Hons.)

Computer Science and Engineering

By

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Institute of Engineering & Technology

GLA University

Mathura- 281406, INDIA

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(TEAM 18)

JATIN KHETAN 2315800039

PARTH GARG 2315800060

ARJUN SINGH RAJPUT 2315800015

Under mentorship of

Prof. Yunis Ahmed Lone

Mentor's Signature

06/02/26

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ABSTRACT

Indoor navigation remains a challenging problem due to the unavailability of GPS signals within buildings and the complexity of indoor environments such as university campuses, hospitals, and commercial complexes. Traditional indoor navigation solutions often rely on infrastructure-heavy technologies like Bluetooth beacons, Wi-Fi fingerprinting, or physical markers, which increase deployment cost and limit scalability. Moreover, conventional map-based navigation interfaces are often unintuitive for first-time users.

The **AR-NAV 2D** project proposes a **vision-based indoor navigation system** enhanced with a **2D Augmented Reality (AR) interface** to address these limitations. The system utilizes architectural floor maps as input, processes them using computer vision techniques to identify walkable regions and obstacles, and applies the **A*** pathfinding algorithm to compute the shortest navigable route between selected points. To improve realism and usability, the computed paths are smoothed and converted into human-friendly trajectories.

The navigation guidance is presented through a camera-based AR overlay, where directional arrows and turn-by-turn instructions are rendered directly on the smartphone's live camera feed. This approach provides an intuitive and immersive navigation experience without requiring any additional physical infrastructure. The system is implemented using **Unity**, **AR Foundation**, and **ARCore**, and is designed as an Android-based prototype suitable for single-floor indoor environments.

The **primary objective of AR-NAV 2D** is to demonstrate an affordable, easily deployable, and user-centric indoor navigation solution for academic institutions. By combining path planning algorithms with augmented reality visualization, the project highlights how AR can significantly enhance indoor wayfinding while maintaining simplicity, flexibility, and scalability. The system serves as a strong foundation for future enhancements such as multi-floor navigation, automated localization, and advanced AR-based guidance.

CONTENTS

1. Overview 5
   - Problem Statement: 5
   - Goal 5
   - Non-goals 6
   - Value Proposition 6
2. Scope and Control 7
   - 2.1 In-Scope 7
   - 2.2 Out-of-Scope 7
   - 2.3 Assumptions 7
   - 2.4 Constraints 7
3. Stakeholders and RACI 8
4. Team and Roles 9
5. Week-wise Plan and Assignments 9
6. Users and UX 10
   - 6.1. Personas 10
   - 6.2. Top User Journey 10
8. Objectives and Success Metrics 13
11. Data Design 16
13. Quality: NFRs and Testing 18
14. Delivery and Operations 19
15. Risks and Mitigations 20
16. Research and Evaluation 21
17. Appendices 22

Revision History

Version	Date	Author	Change
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v1.0	5th Feb, 2026	Jatin Khetan	Formatted to Perfection

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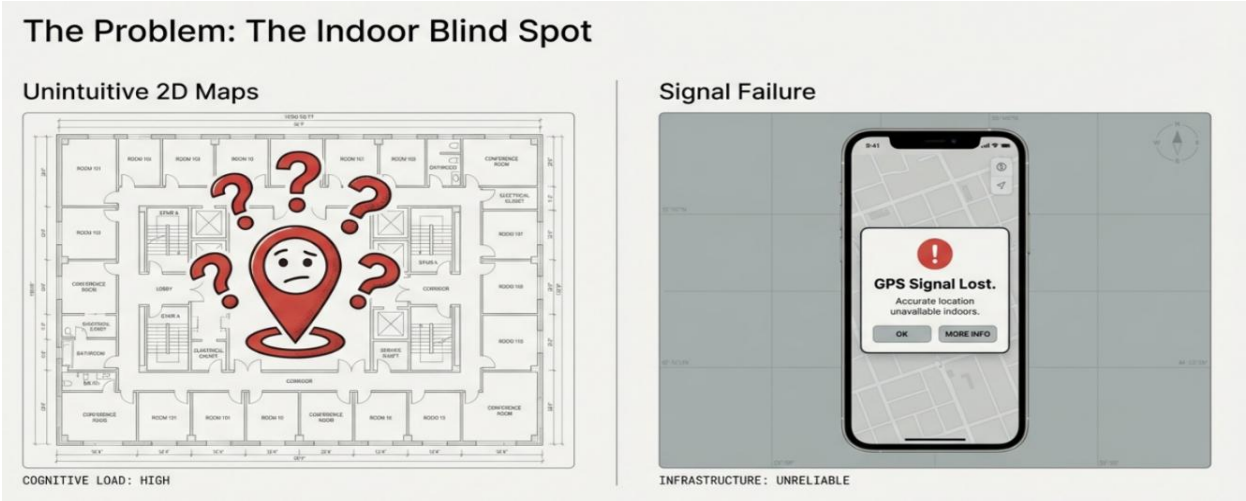
1. Overview

This document defines the scope, architecture, and execution plan for the AR-NAV 3D project. It articulates the core problem of indoor navigation, details our proposed technical solution, and establishes the strategic value of developing a self-hosted, scalable system for the university campus.

- **Problem Statement:** Global Positioning Systems (GPS) are unreliable indoors, failing completely within large, complex structures like university campuses. Existing indoor navigation solutions often depend on expensive, infrastructure-heavy technologies such as Bluetooth beacons or proprietary cloud-based spatial anchors. These systems incur high setup and maintenance costs, require continuous licensing fees, and lack the flexibility for a self-hosted, owner-centric deployment suitable for an academic institution.

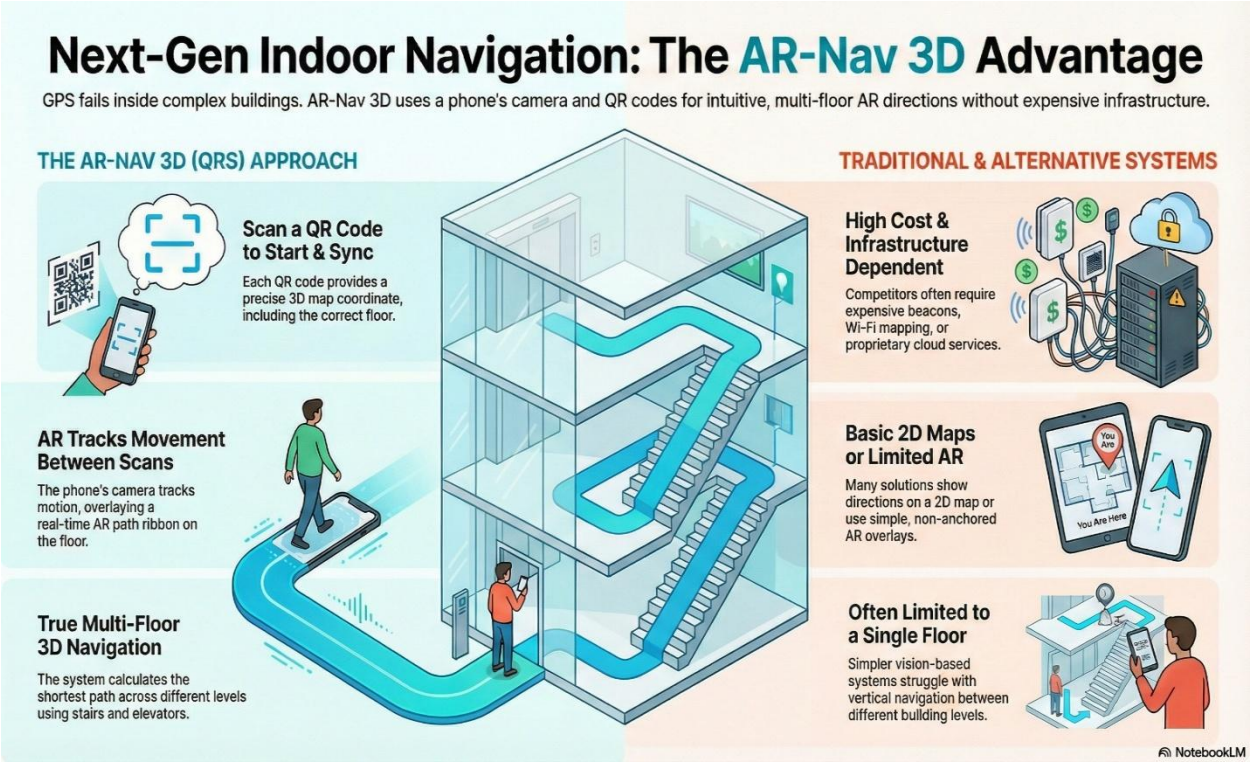


- **Goal:** The primary objective of this project is to design, develop, and deploy **AR-NAV 3D**, a fully functional Augmented Reality indoor navigation system for multi-floor buildings. The system will leverage a QR-based positioning system (QRS) to provide accurate, low-cost,

and intuitive turn-by-turn guidance within a designated pilot building on the GLA University campus.

- **Non-goals** the initial version of this project will explicitly not include the following capabilities:
  - Real-time dynamic obstacle avoidance (e.g., detecting and routing around people or temporary objects).
  - Fully autonomous indoor positioning using Simultaneous Localization and Mapping (SLAM).
  - Infrastructure-based localization using methods such as Bluetooth Low Energy (BLE) beacons or Wi-Fi fingerprinting.
  - **Value Proposition** The AR-NAV 3D system offers a unique value proposition tailored for university environments. Its core differentiators include a significantly **low infrastructure cost**, requiring only printed QR markers. The system is designed as a **campus-owned and self-hosted architecture**, eliminating dependency on third-party cloud platforms and subscription fees. This approach enables a scalable, **block-by-block rollout** strategy and is architected for operational resilience, with core navigation functioning entirely offline—a critical advantage in campus environments with unreliable network coverage.

    This overview establishes the foundation for the project, which is further detailed in the scope definition that follows.



2. Scope and Control

Successful execution demands clearly defined project boundaries. This section delineates the specific features and functionalities within the project's scope, those explicitly excluded, and the underlying assumptions and constraints that govern its development and deployment.

2.1 In-Scope

- **Pilot Deployment:** The system will be deployed in a single, high-traffic university building to validate its functionality and user experience.
- **QR Code-Based Localization:** The core positioning mechanism will use strategically placed QR codes as physical anchors to initialize and recalibrate the user's location.
- **VIO Motion Tracking:** The system will utilize ARCore/ARKit's Visual Inertial Odometry (VIO) to track the user's movement and orientation between QR code scans.
- **3D Pathfinding:** An A* search algorithm will be implemented to compute the optimal path across multiple floors on a pre-defined 3D graph of the building.
- **AR Path Visualization:** A real-time AR "path ribbon" will be rendered on the user's device screen, overlaid on the camera feed to provide intuitive, step-by-step visual guidance.

2.2 Out-of-Scope

- **Campus-Wide Deployment:** The initial project phase is limited to a single pilot building and will not extend to the entire campus.
- **Real-time Crowd Detection:** The system will not detect or account for dynamic obstacles such as crowds of people.
- **Integration with External Systems:** There will be no integration with third-party mapping services like Google Maps.

2.3 Assumptions

- **Accurate Floor Plans:** The project assumes the availability of accurate building blueprints and floor plans to create precise 3D models.
- **User Device Compatibility:** Users are assumed to possess modern smartphones compatible with ARCore (Android) or ARKit (iOS).
- **Static Environment:** The physical layout of the building (walls, corridors, stairwells) is assumed to remain static during a navigation session.

2.4 Constraints

- **Engine Dependency:** The project is constrained to development within the Unity 3D engine and its associated AR Foundation framework.
- **Manual Asset Creation:** Our process requires manual creation of the 3D digital twin models of the building interior using modeling software like Blender.

3. Stakeholders and RACI

To ensure clear ownership and effective communication, this project utilizes a RACI matrix to define stakeholder roles and responsibilities. The matrix clarifies who is **Responsible** for executing a task, who is **Accountable** for its completion, who must be **Consulted** for input, and who will be kept **Informed** of progress.

Activity	Project Lead	AR/Unity Developer	Algorithm Specialist	Faculty Mentor	Department Head
Requirements Definition	A	C	C	C	I
System Architecture	A	R	R	C	I
3D Environment Modeling	A	R	I	C	I
AR & Unity Development	A	R	C	C	I
Pathfinding Algorithm Implementation	A	C	R	C	I
Final Testing & Deployment	A	R	R	C	I

4. Team and Roles

The successful execution of the AR-NAV 3D project depends on a skilled and well-coordinated team. This section outlines the core project members, their designated roles, and their key responsibilities, ensuring that all critical aspects of development are covered by a designated owner.

Member	Role	Responsibilities	Key skills	Contact
Jatin Khetan	Product Lead	Scope, backlog, reviews	Product, APIs	Jatin.khetan_cs.h23@gla.ac.in
Arjun Singh Rajput	Tech Lead & Backend	Arch, APIs, security	Node, Express, SQL	Arjun.rajput_cs.h23@gla.ac.in
Parth Garg	Frontend	React UI, state, a11y	React, TS	Parth.garg_cs.h23@gla.ac.in

5. Week-wise Plan and Assignments

Our execution is structured around a rigorous 8-week plan. The following schedule delineates critical milestones, ensuring a disciplined progression from design and asset creation through to integration, testing, and final demonstration.

Week	Milestone	Key Activities
Weeks 1-2	**Requirements & Design**	Finalize project scope and success metrics. Collect building floor plans. Design the 3D indoor graph structure and QR code data schema. Define system architecture.
Weeks 3-4	**Asset Creation & Algorithm Dev**	Model the pilot building's interior as a low-polygon digital twin in Blender. Implement and unit test the A* pathfinding algorithm logic.
Weeks 5-6	**Unity & AR Integration**	Integrate 3D models into the Unity project. Implement QR code scanning and VIO-based motion tracking. Connect the A* pathfinding engine to the AR visualization layer.
Week 7	**Testing & Refinement**	Print and deploy QR codes throughout the pilot building. Conduct User Acceptance Testing (UAT) with target personas. Identify and resolve critical bugs.
Week 8	**Final Report & Demonstration**	Prepare the final project synopsis and documentation. Record a comprehensive demo video showcasing the end-to-end user journey. Prepare for the final project presentation.

6. Users and UX

A successful navigation system is defined by its ability to provide a clear, intuitive, and seamless user experience. This section identifies the target users for the AR-NAV 3D system and outlines their primary interaction journey, from launching the application to reaching their destination.

6.1. Personas

- **New Student:** A first-year undergraduate who is unfamiliar with the campus layout and needs to find a specific classroom, lab, or lecture hall within a complex, multi-floor academic building.
- **Campus Visitor:** A guest lecturer, conference attendee, or parent attending an event who needs to locate a faculty office, administrative department, or event venue without prior knowledge of the building.

6.2. Top User Journey

The primary user journey for navigating with the AR-NAV 3D system is as follows:

1. Initiation: The user arrives at a building entrance and scans the main "Start Navigation" QR poster with their smartphone.

2. Destination Selection: The application loads the corresponding building map and presents a user interface to select or search for a destination (e.g., room number, lab name, or faculty office).

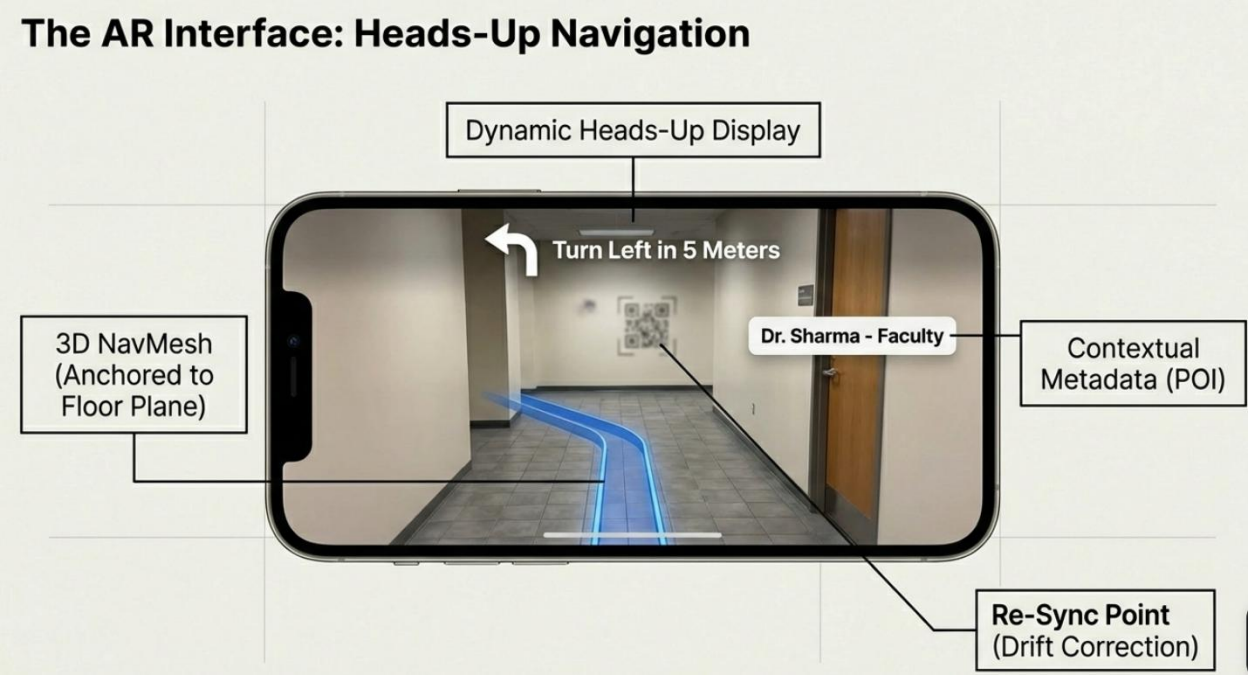
3. Path Computation: Upon selection, the system's A* engine computes the optimal multi-floor path from the user's current location to the destination.

4. AR Guidance: A virtual path ribbon is rendered in real-time and overlaid on the user's camera view, appearing anchored to the floor and indicating the direction of travel.

5. Navigation & Recalibration: The user follows the AR path. At key decision points like corridor junctions or new floors, they scan intermediate QR codes to correct any accumulated VIO drift and re-synchronize their position with the map.

6. Arrival: The system provides a clear notification (visual and/or haptic) when the user has arrived at their destination.

This user-centric journey is enabled by the technical architecture detailed in the following sections.



7. Market and Competitors

Understanding the competitive landscape is crucial for positioning the AR-NAV 3D project and highlighting its strategic advantages. This section evaluates existing indoor navigation solutions to underscore the unique differentiators of the QR-based Positioning System (QRS) approach.

Competitor	Core Technology	Key Weakness / Limitation	AR-NAV 3D Differentiator
Google Maps (Indoor Live View)	Proprietary visual positioning	Limited venue availability; not an owner-centric platform for custom campus mapping.	**Campus-owned and self-hosted:** Provides full control over indoor maps and data.
Pointr	Beacons, sensors ("Deep Location")	Requires expensive infrastructure installation, integration services, and recurring licensing fees.	**Zero infrastructure cost:** Relies solely on inexpensive, printed QR markers.
ARway.ai	Visual markers tied to a SaaS platform	Creates dependency on a third-party cloud platform and its associated pricing model.	**No external dependency:** Core navigation functions without reliance on external clouds or subscription fees.
MazeMap	2D/2.5D web maps	Augmented Reality is a secondary feature; primary focus is on traditional 2D map navigation.	**AR-first experience:** Focuses on true multi-floor 3D AR navigation with a real-time path ribbon.

In summary, the AR-NAV 3D's QRS methodology is a strong fit for university environments, offering a low-cost, scalable, and self-sufficient solution that prioritizes an immersive AR experience.

8. Objectives and Success Metrics

To ensure a rigorous evaluation of the project's success, this section defines specific, measurable, and technical objectives. Each objective is paired with a Key Performance Indicator (KPI) that will be used to validate the system's performance and usability against pre-defined targets.

8.1. **Localization Accuracy:** The system must accurately determine the user's position upon scanning a QR code and maintain minimal drift during movement between markers.

- **KPI:** VIO-induced positional drift must be less than 2 meters over a 30-meter path between QR code re-scans.

8.2. **Pathfinding Efficiency:** The A* algorithm must compute optimal routes on the 3D graph in real-time without introducing noticeable delay for the user.

- **KPI:** Path computation time from any start point to any end point within the pilot building must be under 500 milliseconds.

8.3. **Navigation Usability:** End-users must be able to successfully and intuitively navigate to their selected destination using the AR interface.

- **KPI:** Achieve a task completion rate greater than 90% during controlled user acceptance testing scenarios within the pilot building.

9. Key Features

This section breaks down the AR-NAV 3D system into its core functional components. Each feature has been prioritized to ensure that development efforts are focused on delivering a minimum viable product that fulfills the project's primary objectives.

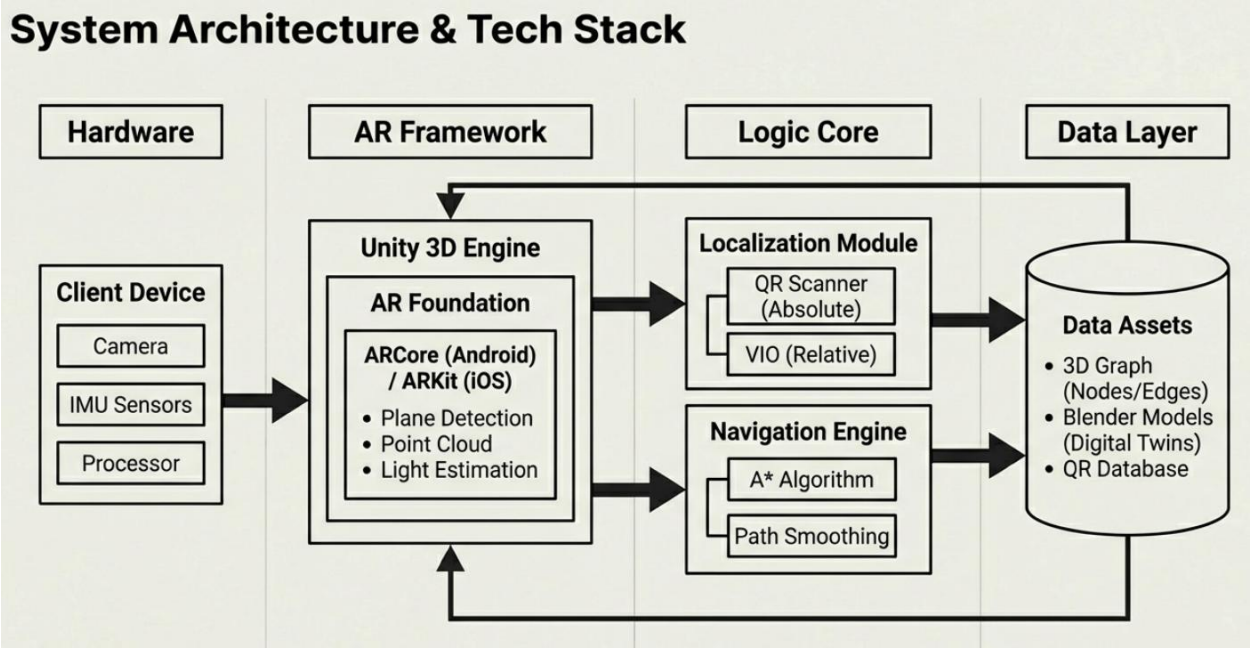
Feature	Description	Priority
QR-Based Localization (QRS)	The use of QR codes as physical anchors encoding 3D coordinates (position and orientation) to initialize and recalibrate the user's position within the building.	Must
VIO-Based Motion Tracking	The utilization of ARCore/ARKit's Visual Inertial Odometry to track the user's relative movement and orientation in the spaces between QR scans.	Must
Multi-Floor 3D Pathfinding	The implementation of an A* search algorithm on a 3D indoor graph to calculate the shortest path, including vertical traversal via stairs and elevators.	Must
AR Path Visualization	The real-time rendering of a visual path ribbon overlaid on the smartphone's camera feed to provide clear, intuitive, step-by-step guidance to the user.	Must
AR Informational Nodes	The display of contextual 3D "info bubbles" with metadata (e.g., room name, faculty designation) when the camera is pointed at pre-defined points of interest.	Must

10.Architecture

This section details the system's technical architecture, its primary components, and the technology stack selected for implementation.

10.1 High-Level Architecture

- **Client Application:** A mobile application built in the Unity 3D engine, utilizing the AR Foundation framework for cross-platform (Android/iOS) AR functionality.
- **Localization Module:** A hybrid system that combines a QR code scanner for absolute positioning and initialization with ARCore/ARKit VIO for continuous relative motion tracking.
- **Pathfinding Engine:** An efficient C# implementation of the A* search algorithm that operates on a pre-defined 3D graph representing the building's navigable spaces, nodes, and edges.
- **3D Environment:** A set of low-polygon digital twin models of the building interiors, created in Blender and imported into the Unity scene to represent the physical space.
- **Data Store (Optional):** A simple backend service (e.g., Firebase/Firestore or a REST API) to manage and serve Point of Interest (POI) metadata for the AR informational nodes.



10.2. Tech Stack

This architecture is supported by a set of well-defined data structures that power the localization and navigation logic.

11.Data Design

The reliability and accuracy of the AR-NAV 3D system are underpinned by two critical data structures: the QR code data schema and the 3D indoor graph model. This section details the design of each.

11.1. QR Code Data Schema

Each QR marker is a physical anchor that encodes a structured data payload. When scanned, this data allows the application to instantly localize the user. The key data fields are:

- **BlockID:** A string identifier for the building (e.g., 'BLKIX').
- **FloorID:** An integer or string for the floor number (e.g., 'F1').
- **Global Coordinates:** A vector representing the precise X, Y, Z position of the marker within the building's global coordinate system.
- **Orientation:** A quaternion or set of Euler angles representing the facing direction required for initial AR session alignment.
- **NodeID:** The unique identifier of the corresponding node in the pathfinding graph (e.g., 'BLKIX_F1_JN03'), linking the physical marker to the logical map.

11.2. 3D Indoor Graph Structure

The pathfinding logic operates on a logical 3D graph that represents all navigable routes within the building:

- **Nodes:** These represent key decision points or locations, such as corridor junctions, room entrances, stair landings, and elevator doors. Each node corresponds to a physical location.
- **Edges:** These represent walkable paths connecting two nodes. Each edge has a weight, typically corresponding to the physical distance between the nodes.
- **Vertical Connectors:** These are special edges in the graph that connect nodes across different floors, representing traversals via stairs or elevators. These edges can have additional cost weights to model factors like elevator wait times.

12. Technical Workflow Diagram

While visual diagrams are omitted in this format, this section describes the primary technical workflow as a sequential narrative. This sequence details the flow of data and system actions from the initial QR scan to the user's arrival at their destination.

The **User Localization & Navigation Sequence** is as follows:

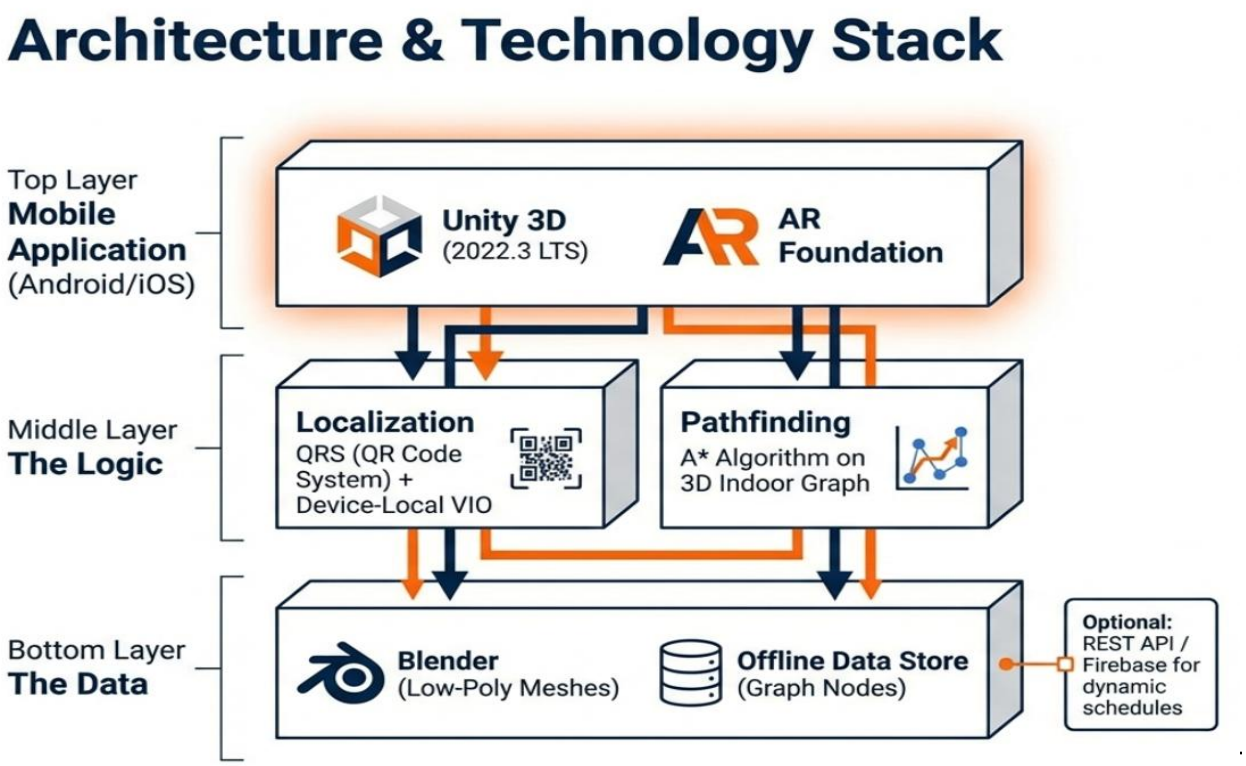
1. Initialization: The user scans an entry QR code. The application decodes its payload to retrieve the NodeID, Global Coordinates, and Orientation.

2. AR Session Alignment: The application's internal AR coordinate system is "snapped" to the real-world position and orientation defined in the QR code. The corresponding 3D building model is loaded and perfectly aligned with the physical environment.

3. Path Request: The user selects a destination from the UI. The system identifies the start NodeID (from the scanned QR code) and the destination NodeID.

4. Calculation: The A* algorithm executes on the 3D graph, finding the shortest sequence of nodes and edges from the start node to the destination node.

5. AR Path Rendering: The resulting path (a sequence of coordinates) is converted into a 3D mesh (the "path ribbon") and rendered within the AR view, appearing visually fixed to the floor.



13.Quality: NFRs and Testing

System quality will be validated against a rigorous set of Non-Functional Requirements (NFRs). This section defines the performance, accuracy, and robustness targets that the AR-NAV 3D system must achieve to be considered successful.

13.1 Non-Functional Requirements

Metric	Target	Measurement Method
Performance	Path calculation latency must be < 500ms.	Measured via internal application logging during test runs.
Accuracy	VIO drift between QR scans must be < 2m over 30m.	Measured via controlled physical walk-throughs using pre-measured distance markers on the floor.
Robustness	QR scanner must correctly read codes in various lighting conditions (down to 200 lux).	Measured in a controlled environment using a lux meter and a standardized set of test QR codes.

13.2 Test Plan

- **Unit Testing:** The A* pathfinding algorithm module will be rigorously tested in isolation with a mock graph data set to verify path correctness, efficiency, and handling of edge cases.
- **Integration Testing:** Tests will be conducted to verify the seamless data flow between the QR scanner module, the localization engine, the A* pathfinding engine, and the AR rendering module.
- **User Acceptance Testing (UAT):** Supervised tests will be performed with target users (students and visitors) within the fully modeled and QR-tagged pilot building. This will validate the end-to-end user journey and gather qualitative feedback on usability and intuitiveness.

14.Delivery and Operations

This section outlines the project's deployment strategy and the considerations for its long-term operational maintenance.

15.1 Release Plan

The system will be deployed using a phased rollout strategy. The initial release will be a **pilot deployment** confined to a single, high-traffic university building, designated as Block IX. This controlled launch will allow for thorough testing and user feedback collection before any consideration is given to broader expansion.

15.2 CI/CD and Rollback

For this academic project, a manual build and deployment process will be used for the client application. A simple semantic versioning system (e.g., v1.0, v1.1) will be maintained for all builds. In the event of a critical issue discovered post-deployment, this system allows for a straightforward rollback to a previously known stable build.

15.3 Operations and Maintenance

Long-term operation of the AR-NAV 3D system will require the following maintenance activities:

- **QR Code Maintenance:** Periodic physical inspection of all deployed QR markers to check for damage, obstruction, or fading. Damaged markers will be replaced to ensure scanner reliability.
- **Map Updates:** In the event of physical changes to the building's layout (e.g., renovations, new room partitions), the corresponding 3D models and the pathfinding graph must be updated and redeployed.

15.Risks and Mitigations

Proactive risk management is essential for anticipating and addressing potential challenges. This section identifies the key technical and logistical risks to the AR-NAV 3D project and outlines corresponding mitigation strategies to minimize their impact.

Risk	Probability	Impact	Mitigation Strategy
VIO Drift Accumulation	High	Medium	Strategically place QR codes at all decision points (corridor junctions, stairs, elevators) to ensure frequent re-syncing opportunities, thereby resetting any accumulated VIO drift before it impacts user experience.
Inaccurate Floor Plans	Medium	High	Verify the dimensions obtained from official building blueprints with on-site physical measurements before finalizing the 3D models. This ensures the digital twin accurately reflects the real-world environment.
Poor QR Code Readability	Medium	Medium	Use high-contrast printing on non-reflective material to create the physical markers. Place codes in well-lit areas away from potential glare. Implement error correction capabilities within the QR scanner logic.

16. Research and Evaluation

This project is built upon established research and will be evaluated through a structured plan to measure its success and identify areas for future improvement.

Research Basis: The project's core methodology is supported by existing academic research and prototypes which demonstrate that QR-based Augmented Reality indoor navigation systems are a viable, accurate, and user-friendly solution, particularly for campus and library environments.

Evaluation Plan: The success of the AR-NAV 3D system will be evaluated by executing the User Acceptance Testing (UAT) scenarios outlined in the Test Plan (§13.2). Following the tests, user feedback will be collected through structured surveys to quantitatively and qualitatively measure system usability, accuracy, and overall user satisfaction against the objectives defined in §8.

Limitations & Future Work: The current version of the project has several acknowledged limitations, which also define a roadmap for future enhancements.

- **Current Limitations:**
  - Localization is entirely dependent on the presence of physical QR markers.
  - The system relies on a static environment assumption and does not account for dynamic obstacles like people or temporary furniture.

- **Future Work:**
  - Integrate visual SLAM (Simultaneous Localization and Mapping) to eventually phase out the dependency on QR markers for localization, moving towards a truly infrastructure-free system.
  - Implement dynamic obstacle detection and real-time path rerouting.
  - Add voice-based navigation commands for an accessible, hands-free user experience.

17. Appendices

This section contains supplementary information, including a glossary of technical terms and references.

- **Glossary**
  - **Augmented Reality (AR):**
    A technology that overlays digital information such as graphics, text, or directions onto the real-world environment through a device camera.
  - **Indoor Navigation:**
    The process of guiding users within enclosed environments such as buildings where GPS signals are unreliable or unavailable.
  - **A* (A-Star) Algorithm:**
    A heuristic-based graph search algorithm used to compute the shortest and most efficient path between two points.
  - **Pathfinding:**
    The computational process of determining an optimal route between a start and destination while avoiding obstacles.
  - **Computer Vision:**
    A field of artificial intelligence that enables computers to interpret and process visual information from images or videos.
  - **Floor Map:**
    A 2D architectural representation of an indoor environment used to identify rooms, corridors, and obstacles.
  - **Walkable Area:**
    Regions within a floor map where user movement is permitted, excluding walls, furniture, and restricted zones.
  - **Obstacle Detection:**
    The identification of non-navigable regions such as walls or objects that block movement.
  - **Path Smoothing:**
    A technique used to refine grid-based paths into continuous, human-friendly navigation trajectories.
  - **Turn-by-Turn Navigation:**
    A navigation method that provides sequential instructions such as “go straight” or “turn left” based on the user’s movement.
  - **AR Overlay:**
    A visual layer displayed over a live camera feed to present navigation cues and instructions.
  - **Unity:**
    A cross-platform game engine used to develop interactive 2D, 3D, and AR applications.
  - **ARFoundation:**
    A Unity framework that enables the development of augmented reality applications across multiple platforms.
  - **ARCore:**
    Google’s platform for building augmented reality experiences on Android devices.

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